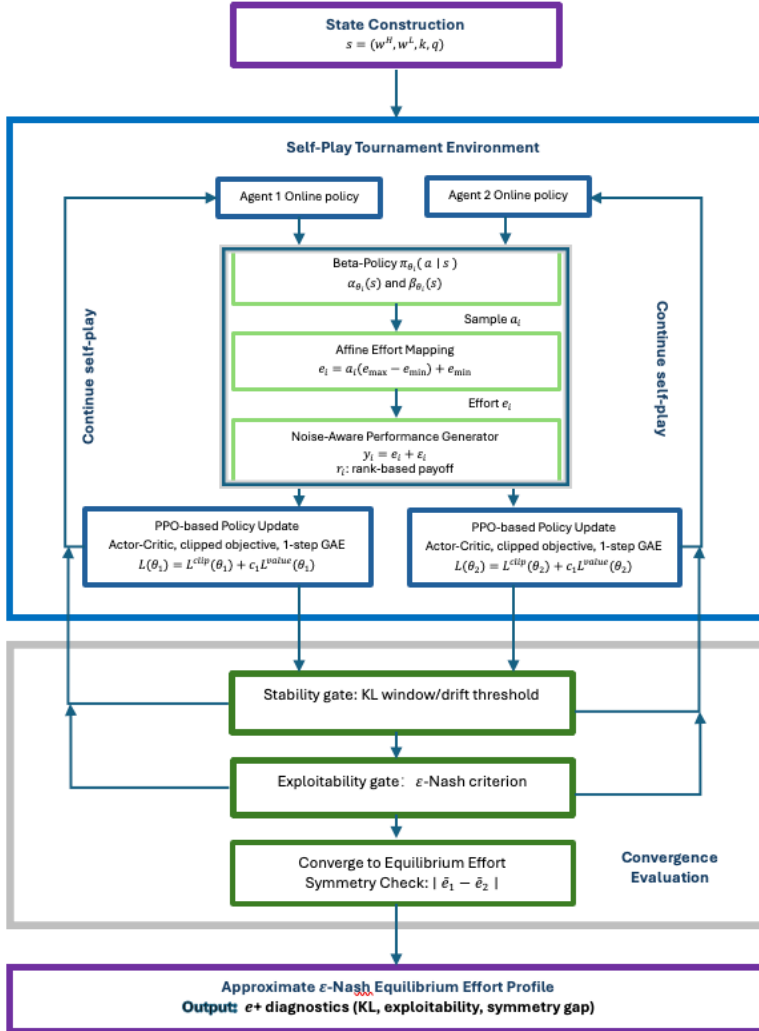


One-stage Tournament

Figures:

1- Figure 1: Tel-PPO Framework

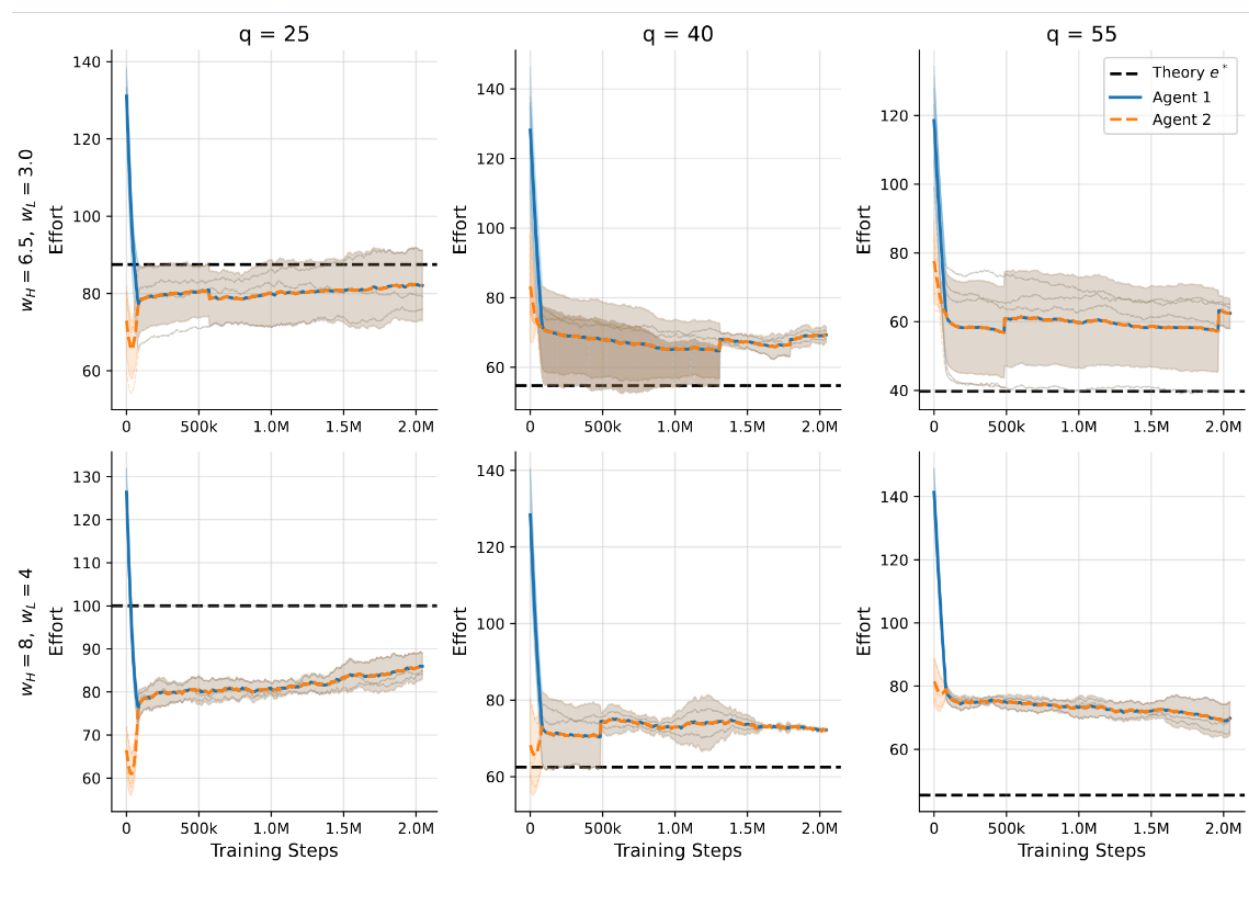


Static tournament primitives (degenerate one-step MDP); No state transitions across time; Independent agents with identical architecture (no parameter sharing); Continuous bounded action space via Beta distribution; Synchronous updates under self-play; Monte Carlo best-response search (common random numbers); Declared when criteria hold for consecutive evaluations; Diagnostic in symmetric settings only

2- Figure 2. Two-player baseline under varying noise q :

a- Effort trajectory vs training (two sets of k & w value)

x-axis: training steps; y-axis: effort [0, 200]; two agents' efforts: e_1, e_2 , dashed line: theory e^* ; dotted line: $\frac{e_1 + e_2}{2}$; panel labels: noise level $q =$; Convergence detection is indicated by a vertical line (or shaded region)?

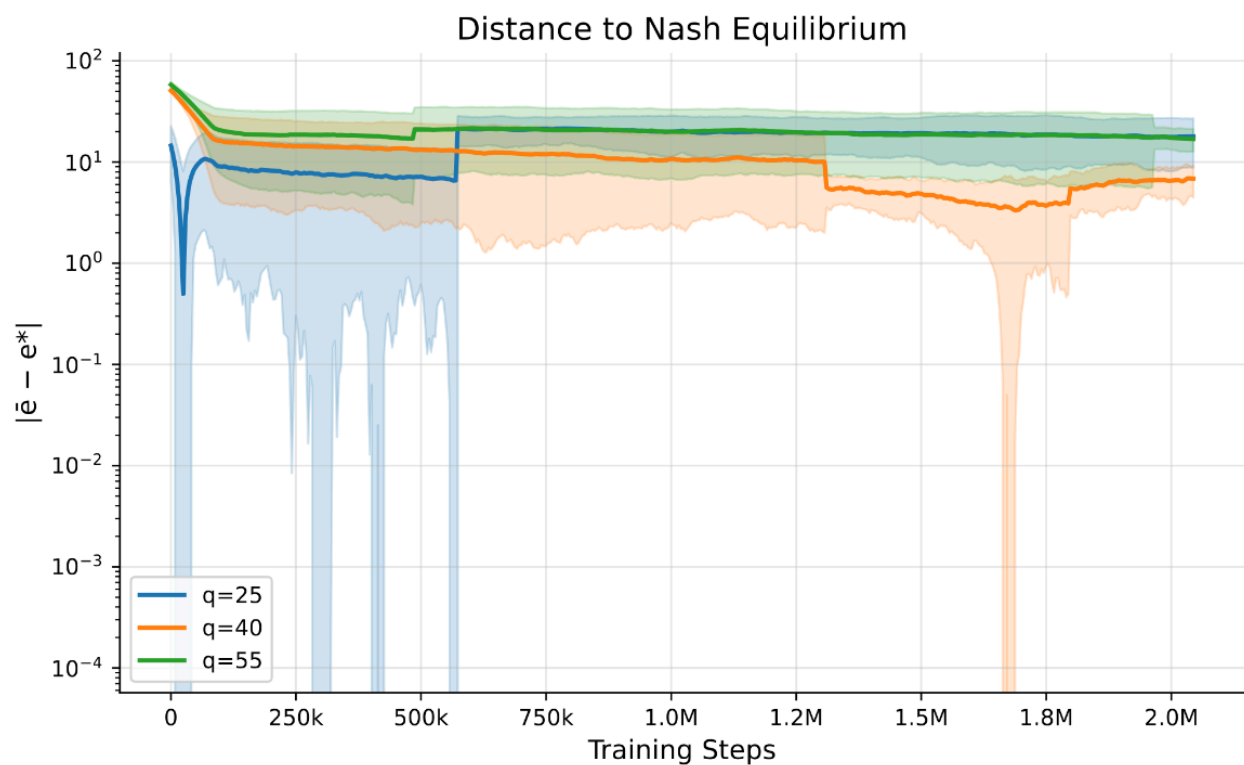


Revision ideas:

- 加一条 Convergence detection 线;
- 每个 panel 右上角加一个 Final Summary: Final $|\bar{e} - e^*| = ?$, Exploitability = ? ($\epsilon = ?$), symmetry gap (或者标注下来, 我之后加);
- theory 换个颜色看一下效果 (红色粗线?);
- 降低阴影透明度;
- 统一 y 轴;
- 加上 Noise Level $q=25/40/55$;
- 选用 $\text{mean} \pm \text{CI95}$ 有什么特殊的考虑吗? 换成 std 的效果怎么样?
- 现在的阴影是多少个 seeds 的?

b- Distance-to-equilibrium

x-axis: training steps; y-axis: $|\bar{e} - e^*|$, different noise q



Revision ideas:

a- y 轴从 10^{-1} 开始?

b- 体现 convergence step, 可以用同色竖线标注;

c- 加一条 line 表示 ε threshold;

d- 改成 mean+-std 会不会减少 downward spikes

e- 降低阴影透明度

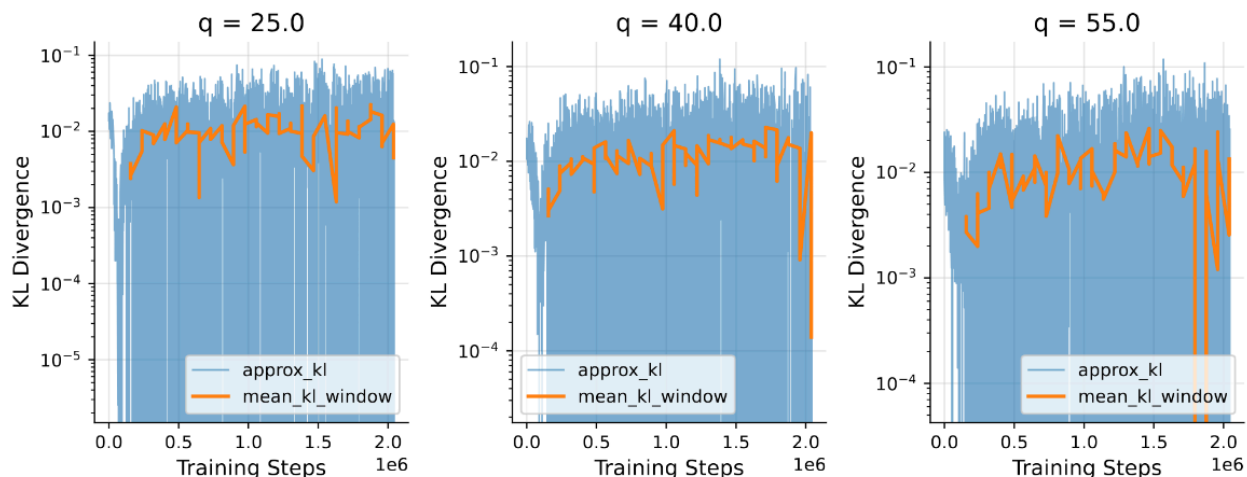
e- shaded 部分是 across 不同的 seeds 吗? 多少个 seed

f- Name: Distance to the Nash Equilibrium Across Noise Levels

3- Figure 3: Training stability diagnostics

a. Policy drift (KL)

x-axis: training steps; y-axis: approx. KL; line: target KL or thresh; shadow: rolling window



Revision ideas:

a- 加上 target KL/cheap-gate mean KL threshold;

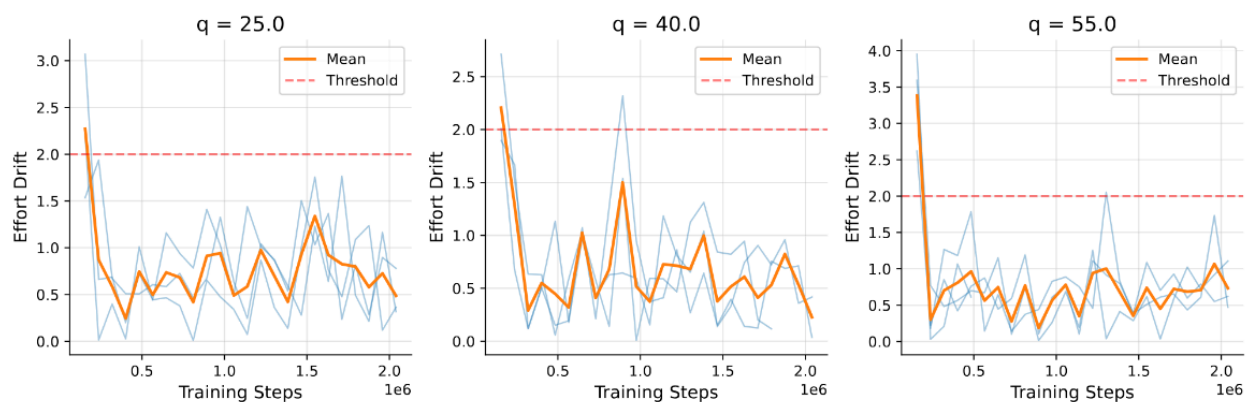
b- 每个 seed 的 approx_kl 用细线且高透明度，或者每隔 10-50 个点画一个；如果有多个 seed，用 mean approx_kl 的效果是不是会好些；

c- 设三个 panel 的 y-limits [$1e-4$, $1e-1$] 效果会不会好些

d- Is window size 20? I will mention it in caption.

b. Effort drift

x-axis: training steps; y-axis: $|\bar{e}(t) - \bar{e}(t - \Delta)|$ (Δ = window length); line: drift threshold, window size, mean KL thresh / drift thresh



Revision ideas:

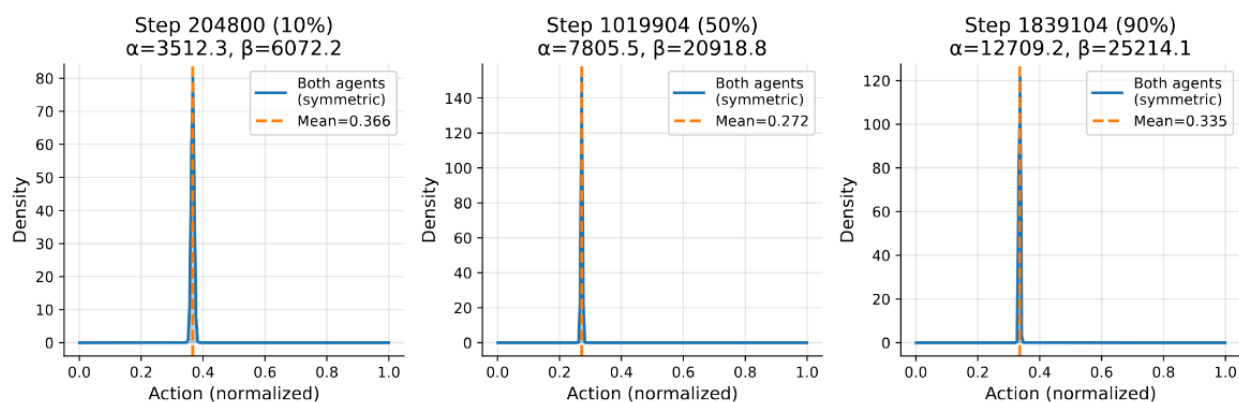
a- 统一 scale;

b- threshold 加粗或者换个颜色，并标注数值或者我放到 caption 里

c- 标注蓝线是每个 seed 的 trace. 有几个？我可以标注在 caption 里

4- Evolution of Beta policy distribution (two player baseline)

x-axis: effort; y-axis: density; three checkpoints: 10%, 50%, 90%; two beta distribution with mean and variance



Revision ideas:

a- 统一 scale;

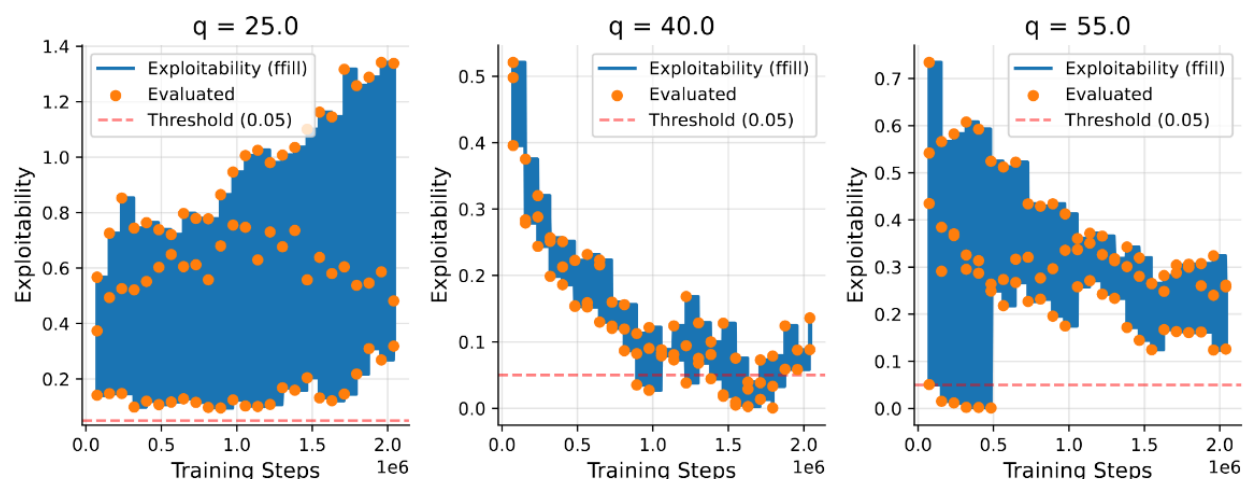
b- 加一条其他颜色的线标注 c^* 值;

c- use $k=\alpha+\beta$ as concentration parameter, 并缩小字

d- 两个 agent 的 beta 曲线用不同颜色表示叠加

5- Figure 5: Exploitability curve / Equilibrium verification

x-axis: training steps; y-axis: exploitability max; horizontal line ε ; evaluation frequency (how many PPO updates)



Revision ideas:

a- 统一 y-axis

b- log y-scale 0.01-1 或许可以更清晰显示是否接近 threshold

b- threshold 加粗更明显一些;

c- 加一个 convergence 竖线标记,

d- 每个 seed exploitability 尝试改成细线, 增加颜色透明度; 推荐用 step plot 体现 ffill value

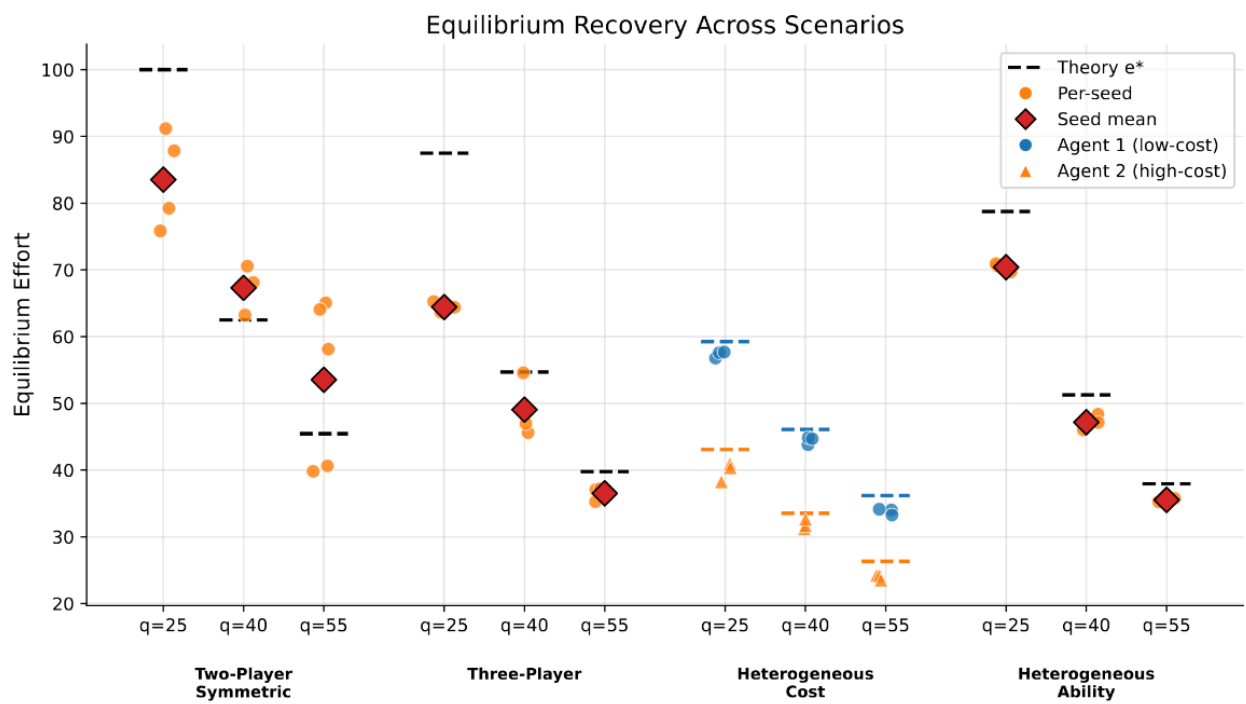
6- Equilibrium Recovery Across Tournament Structures: Dot Plot with theory line

x-axis: scenario (Two-player, Three-player, Heterogenous ability, heterogeneous cost)

y-axis: learned equilibrium effort; theory line.

TEL-PPO recovered effort: symmetric case-one dot (mean effort); heterogeneous cost-one dot for low-cost agent, one triangle for high-cost agent (e_1, e_2)

Theory e^* : short line



Revision ideas:

a- two-player 的结果是 $k=0.0005$, $w^h=8$, $w^l=3$ 那组值的结果是吗? baseline 的结果是不是更好?

b- 每一组 seed mean 的相对误差如何? 如果有说服力的话可以考虑放在 caption

b- theory 线更明显一些, 加粗

Table:

1- Environment configuration

Parameter	Value/Description
(High prize w_H , Low prize w_L)	(6.5,3), (8,4)
Cost parameter	$k = 0.0004$
Action/Effort bounds	$[e_{\min}, e_{\max}]$: [0,200]
Noise distribution	$\varepsilon_i \sim U[-25,25], U[-40,40], U[-55,55]$
Hyperparameter	Learning rate, batch size, clip, seeds? held constant across experiments unless stated?

2- Quantitative summary of equilibrium recovery

Scenario	q	Method	Mean $\pm$ std?	$ \bar{e} - e^* $	Exploitability	Symmetry gap	Steps to convergence
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3- Ablation

Variant	TEL-PPO	+Entropy	No stability gate	No exploitability gate
Final error				
exploitability				
Steps to convergence				
Failure mode				

Failure mode: diverge/ cycle / biased mean / never terminate